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Behavioral and Adrenocortical Responses of Rhesus Macaque Mothers to Infant Separation in an Unfamiliar Environment

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ABSTRACT. The behavioral and adrenocortical responses of rhesus macaque mothers to a series of four consecutive 4-day separations from their 5-month old infants in an unfamiliar environment were examined. A biphasic behavioral response to separation was observed, with passive behavior, locomotion, and vocalization highest on Day 1 of separation, and self-directed behaviors and environmental exploration peaking during Days 2–4. Stereotyped locomotion increased, and passive behavior decreased, across successive weeks of separation. The rhesus mothers exhibited substantial cortisol elevations one and two hours postseparation, with cortisol decreasing on the fourth day of separation. The cortisol response was strongest during the first week of separation, but robust elevations occurred in response to repeated separations. No signs of behavioral depression were observed. The results indicate that infant separation combined with removal to a novel environment can be a potent stressor for rhesus macaque mothers.

Key Words: *Macaca mulatta*; Separation; Behavior; Cortisol; Mother-infant.

INTRODUCTION

The effects of separation from an attachment object have been extensively studied in nonhuman primate infants. Whereas a number of factors such as rearing condition, nature of the separation environment, and species can influence the form and the severity of the separation reaction, the characteristic response involves behavioral agitation concurrent with adrenocortical activation and perturbations of thermoregulation and the sleep cycle (REITE et al., 1974; MINEKA & SUOMI, 1978; COE, WIENER, & LEVINE, 1983). In infants undergoing a prolonged separation, the initial behavioral response diminishes over time, and behavior eventually returns to preseparation levels (MINEKA & SUOMI, 1978). Some infants may lapse into a behavioral state classically referred to as "despair," which can incorporate hunched posture, prolonged self-directed behavior, immobility, refusal of food, and loss of interest in play. In infants, across repeated separations the acute behavioral response to separation declines, whereas behaviors traditionally considered despair behaviors, such as self-clasp and huddle, increase (SUOMI et al., 1970; COE, GLASS, WIENER, & LEVINE, 1983).

The maternal response to separation from infant has been less well investigated. Studies of several species have reported that the immediate response of the mother to separation from her infant is behavioral agitation (JENSEN & TOLMAN, 1962; SEAY et al., 1962; KAPLAN, 1970; HINDE & DAVIES, 1972; VOGT & LEVINE, 1980; SUOMI et al., 1983). Cortisol elevation immediately following infant separation occurs in individually and group-housed squirrel monkeys (MENDOZA et al., 1978; COE et al., 1978). In rhesus, postseparation cortisol elevation occurs only in the absence of the infant (SMOTHERMAN et al., 1979). Rhesus mothers separated from, but allowed visual contact with, their infants and social

group for three hours exhibit cortisol levels comparable to mothers separated with their infants from their social group in view of the group (SMOTHERMAN et al., 1979). However, rhesus mothers whose infants are removed show cortisol elevations one and three hours postseparation (LEVINE et al., 1985). Primate mothers respond to long-term separations with gradual decreases in agitation, with few manifestations of typical "despair" behaviors (JENSEN, 1968; KAPLAN, 1970; SUOMI et al., 1983). In rhesus mothers, SUOMI et al. (1983) reported no evidence for diminished protest behavior across successive separations. In addition, there is no evidence for adaptation of the adrenocortical response to separation in squirrel monkey mothers (COE, GLASS, WIENER, & LEVINE, 1983); similar data do not exist in the published literature for the rhesus.

The present study examined the effects of repetitive infant separation on behavior and adrenocortical responses in dyad-housed rhesus macaque mothers. The separation paradigm consisted of four 4-day separations and was similar to that used by SUOMI et al. (1983) with two exceptions: (1) mothers were placed into an unfamiliar environment during the separation period; and (2) blood samples were collected to assess the adrenocortical reaction to separation. The purpose of the study was threefold:

(1) The nature of the separation environment has been demonstrated to influence the form of the separation response. One important factor is whether the separated animal remains in its home cage or is placed into a novel environment. Infant rhesus monkeys separated into a novel environment display more agitation than those remaining in the home cage during separation (HINDE & DAVIES, 1972; HINDE & MCGINNIS, 1977). A prior study (SUOMI et al., 1983) demonstrated that mothers that remained in their home cages after their infants were removed exhibited mostly inactive behavior. Therefore, we hypothesized that separation of mothers into a novel environment would evoke more maternal distress than leaving the mothers in the home cage, and would enhance any individual differences in the separation response that might exist. With the exception of blood sampling, our study allowed a direct comparison to that of SUOMI et al. (1983).

(2) An additional consideration in assessing the response to social separation is whether the separated animal has access to social partners. In general, provision of social partners tends to buffer the behavioral and adrenocortical response to separation (e.g. GUNNAR et al., 1980). Most studies of mother-infant separation have utilized group-housed monkeys. Under these conditions, comparisons of the separation response in the home cage vs a novel environment can be confounded with presence vs absence of social partners. Only one study has examined separations in dyad-housed rhesus macaque mother-infant pairs (SUOMI et al., 1983). Performing separations on mother-infant dyads was of particular interest to us because there could be no other social partners to buffer the effects of social separation. Using our procedures it is impossible to disentangle the effects of social isolation from those of separation. By definition, any separation of a mother-infant dyad implies social isolation during separation.

(3) Few studies have examined repetitive separations in the monkey mother, and even fewer have incorporated physiological measures. Repetitive behavioral observations and blood sampling allowed us to determine the effects of multiple separations on the adrenocortical and behavioral responses to separation in the mother.

METHODS

SUBJECTS

Subjects were 15 multiparous laboratory-born rhesus macaque mothers ranging in age from 6 – 17 years. Eight females were rearing their biological infants, and seven were rearing unrelated foster infants. Foster infants had been given to their adoptive mothers within one week postparturition, immediately following removal of the biological infant. All mothers had been living with their infants for at least five months at the onset of the study.

Mothers and infants were housed as dyads in 74 × 74 × 74 cm cages in a large colony room containing group-and individually housed monkeys. Visual, olfactory, and auditory, but not tactile, contact was possible with other animals in the room. The room was maintained on a 12:12 lighting schedule. Purina monkey chow was provided on a rationed basis in accordance with standard laboratory protocol; water was available ad lib. Animals were not disturbed except for routine animal care procedures.

PRESEPARATION BEHAVIORAL OBSERVATIONS

Eight 5-min focal animal observations were conducted on mothers in their home cages during the two-week period immediately prior to separation. The observer sat approximately 1 m from the cage, visible to the monkeys. The frequency and duration of 16 social and nonsocial behavioral categories were recorded on a laptop computer. Behavioral definitions of nonsocial behaviors relevant to this study are listed in Table 1.

Table 1. Behavioral scoring categories.

Self-directed: any oral/tactual self contact. During separations this category was broken down into the behaviors:
Self-mouth: any oral contact with self.
Self-clasp: clutching or gripping self.
Self-groom/scratch: grooming or scratching self.
Environmental exploration: tactual and/or oral manipulation of inanimate objects, including eating and drinking.
Locomotor stereotypy: patterned movements maintained in a rhythmic and repetitive fashion for at least three cycles (scored independently of locomotion).
Nonlocomotor stereotypy: patterned movements (which are nonlocomotive in nature) maintained in a rhythmic and repetitive fashion for at least three cycles.
Passive: absence of all social, exploratory, and locomotor activities. May be scored with self mouth/clasp/groom and vocalization only.
Locomotion: any self-induced change in location of self.
Vocalization: any vocalization emitted by the animal.

SEPARATION PROCEDURE

When infants were approximately five months of age, mother-infant pairs were exposed to a repetitive separation paradigm, consisting of four consecutive 4-day separations, interspersed with 3-day reunion periods. Separations were performed at 13:00 on Mondays, and mothers and infants were reunited at 10:00 on Fridays. Separation was performed by one

person manually removing the infant from the mother's ventrum while two other persons restrained the mother manually. Separated mothers were housed in 56 × 61 × 76 cm single cages in an unfamiliar room containing from two to three other separated mothers. The mothers were allowed auditory and olfactory but not visual or tactile contact with each other. Infants were removed to a room where they could be neither seen nor heard by their mothers. Because separations were conducted when infants were approximately five months old, and because the infant birth season extended from April to August, separations were conducted in four consecutive cohorts, containing four, three, four, and four mothers, respectively.

SEPARATION BEHAVIORAL OBSERVATIONS

Behavioral observations were conducted on separated mothers three times on Day 1 of separation (twice in the first hour postseparation, and once in the second hour postseparation), and twice daily between 11:00 and 15:00 during Days 2 – 4. The frequency and duration of ten behavioral categories were recorded as described above during a 5-min observation period. Observations were performed by five testers, trained to a reliability criterion of .95 (Pearson product-moment correlation) averaged across all behavioral categories.

BLOOD COLLECTION PROCEDURE

Blood was sampled approximately one and two hours following each separation, at approximately 14:00 and 15:00. Five ml blood were collected from the saphenous vein, with the conscious animal immobilized in a restraint cage. In this condition, animals were transported in a carrying cage from their separation cage to the restraint apparatus, and were sampled within five min of capture from the home cage. Because the (three or four) separated animals were captured sequentially from within the separation room, it was not always possible to capture all three or four animals within five min following room entry. However, on average, all animals were bled within five min of entry into the room, with a range of sampling times from 1 to 19 min following initial room entry. Analysis of data obtained under this condition indicated no significant correlations between either time of sampling relative to initial room entry, or order of sampling (i.e. first, second, third, fourth) within the room, and cortisol levels.

Blood samples were also obtained on the fourth day of the first and fourth separations, two weeks prior to the first separation, and two weeks following the fourth separation, at approximately 14:00. These blood samples were collected in conjunction with cerebrospinal fluid sampling. During this procedure, mothers were anesthetized with ketamine HCl (10 mg/kg IM) while in their home cages, and allowed to reach unconsciousness in the home cage. The animals were then transported to a test room where blood was obtained after cerebrospinal fluid sampling was completed. Injection of all animals took place within five min following initial room entry. In this condition, the average time of blood sampling following initial room entry was 35.27 min for Preseparation (range 19 – 50 min), 14.29 min for Week 1 Day 4 (range 9 – 20 min), 16.07 min for Week 4 Day 4 (range 5 – 23 min), and 24.64 min for Postseparation (range 12 – 37 min). Animals were weighed following CSF and blood collection during the Preseparation, Week 4 Day 4, and Postseparation time points. All blood samples were spun at 10°C for 15 min and the plasma fraction quick-frozen at −70°. Radioimmunoassay for cortisol was conducted at Hazleton Laboratories, Vienna, Virginia.

Data Analysis

For each behavior scored during Preseparation, the average of the eight Preseparation scoring sessions was obtained and entered into subsequent analyses. For each behavior scored during Separation, an average score was calculated for the three observation periods from Day 1 of separation (Acute phase), and for the six observation periods from Days 2–4 of separation (Chronic phase). The average Acute and Chronic values were then utilized in subsequent analyses. Significance level was set at $p<.05$ for all analyses. All significant main effects or interactions were further analyzed by Duncan post-hoc tests (Duncan, 1955).

Preliminary analyses revealed no significant differences between biological and adoptive mothers on any measure; therefore, these groups were combined for subsequent analyses. Additional preliminary analyses were run correlating mothers' age with behavioral and cortisol measures. The only behavior to exhibit a significant correlation with mothers' age was nonlocomotor stereotypic behavior during separations. However, the data indicate that this is not a meaningful result and reflects one animal's idiosyncratic behavior, for two reasons. First, the significant correlation was primarily due to the oldest mother exhibiting abnormally high levels of nonlocomotor stereotypic behavior. If this female is omitted from the analysis, the correlation becomes nonsignificant. Second, the second oldest female exhibited no nonlocomotor stereotypic behavior during separation.

Comparison of Preseparation and Separation Behavior

Six behaviors from the Preseparation condition could be directly compared with those observed during separation: locomotor stereotypy, nonlocomotor stereotypy, environmental exploration, locomotion, self-directed behavior, and vocalization (passive behavior could not be directly compared between conditions, as passive behavior is inversely related to the amount of social behavior exhibited by the animal). The average scores from the eight Preseparation test sessions were compared to Acute and Chronic phase behaviors by one-way ANOVA with Condition (Preseparation, Acute, and Chronic) as a within-subjects factor.

Separation Behavior Analyses

Within the separation condition, two-way analyses of variance were conducted on the behavioral data, with Week of Separation and Phase of Separation (Acute vs Chronic) as within-subjects factors.

Cortisol Response to Separation Analyses

Several analyses of the cortisol data were conducted to assess different effects. A one-way within-subjects ANOVA was performed comparing cortisol values from the following time points: mean basal value (mean of Preseparation and Postseparation samples), mean Hour 1 (average of Weeks 1 through 4), mean Hour 2 (average of Weeks 1 through 4), and mean Day 4 (average of Weeks 1 and 4). An additional two-way repeated measures ANOVA was conducted on the cortisol values obtained on the Mondays immediately following separations, with the within-subject factors Week of Separation and Hour Postseparation, to allow the assessment of possible habituation of the cortisol response across weeks of separation. The third analysis of variance compared the values obtained during Week 1 of

separation to those of Week 4, with Week of Separation (1, 4) and sample condition (Hour 1, Hour 2, and Day 4) as within-subject factors. The final analysis compared Preseparation to Postseparation cortisol values.

BODY WEIGHT ANALYSES

Body weight data were analyzed using a one-way ANOVA with Time Point (Preseparation, Week 4 Separation, and Postseparation) as a within-subjects factor.

RESULTS

PRESEPARATION VS SEPARATION BEHAVIOR

There were significant effects of Condition for the behaviors self-directed [$F(2,28)=11.12$, $p<.005$], environmental exploration [$F(2,28)=23.69$, $p<.001$], locomotor stereotypy [$F(2,28)=3.75$, $p<.05$], and vocalization [$F(2,28)=20.76$, $p<.001$]. Post-hoc analyses revealed that self-directed behavior was lower in the Acute phase of separation than in either the Chronic or Preseparation conditions ($p<.01$ for both comparisons). Environmental exploration increased from Preseparation to the Acute phase ($p<.01$) and again from the Acute to the Chronic phase ($p<.01$). Locomotor stereotypy in the Acute and Chronic periods were both higher than Preseparation locomotor stereotypy ($p<.05$ by post-hoc analyses for both comparisons). Vocalization was elevated in the Acute phase of separation relative to the Chronic and Preseparation periods (for both comparisons $p<.01$). These results are depicted in Table 2.

Table 2. Preseparation and separation behaviors.

Behavior	Preseparation	Acute phase	Chronic phase
Self-directed	45.4 (8.0)	19.6 (8.0)	55.1 (10.5)
Envt. explore	11.7 (3.6)	33.5 (6.7)	63.8 (8.0)
Loco. stereotypy	2.0 (1.5)	12.9 (5.9)	12.0 (3.5)
Vocalization	2.8 (1.1)	17.1 (3.4)	1.6 (0.5)

All values represent average duration (sec) from 5-min testing sessions, with the exception of vocalization, which represents average frequency. Standard errors in parentheses.

SEPARATION BEHAVIOR

A biphasic response to separation was observed for most behaviors. As shown in Table 3, passive [$F(1,14)=16.39$, $p<.001$], locomotion [$F(1,14)=4.99$, $p<.005$], and vocalization

Table 3. Acute vs chronic phase separation.

Behavior	Acute phase	Chronic phase
Passive	217.0 (10.5)	188.8 (7.8)
Locomotion	22.8 (3.2)	17.2 (2.9)
Vocalization	17.1 (3.4)	1.6 (0.4)
Self-clasp	11.9 (8.4)	23.7 (4.6)
Self-groom	7.0 (1.4)	26.4 (4.6)
Envt. explore	33.5 (6.7)	63.7 (8.0)

All values represent average duration (sec) from 5-min testing sessions, with the exception of vocalization, which represents average frequency. Standard errors in parentheses.

[$F(1,14)=23.02$, $p<.001$] were highest in the Acute phase of separation, whereas self-clasp [$F(1,14)=6.79$, $p<.005$], self-groom [$F(1,14)=19.42$, $p<.001$)], and environmental exploration [$F(1,14)=24.75$, $p<.001$] peaked in the Chronic phase.

There were significant effects of Week of Separation for the behaviors passive [$F(3,42)=3.95$, $p<.05$] and stereotyped locomotion [$F(3,42)=3.20$, $p<.05$]. Passive decreased across weeks; although the Week 1 value was significantly higher than those from Week 3 ($p<.05$) and Week 4 ($p<.01$), no other comparisons reached statistical significance. Stereotyped locomotion increased across successive separations; the only statistically significant difference was between Week 1 and Week 4 ($p<.01$). A significant Week × Phase of Separation effect was noted for self-mouth, $F(3,42)=3.53$, $p<.05$. The Week 1 Chronic phase value of 10.4 sec was higher than all others ($p<.01$ for all possible comparisons); self-mouth in the other test periods was minimal (range from 0.1 − 3.6 sec average), and no other pairwise comparisons were statistically significant. The changes in passive and stereotyped locomotion across weeks of separation are depicted in Table 4.

Table 4. Behavior across weeks of separation.

Behavior	Week 1	Week 2	Week 3	Week 4
Passive	223.1 (10.9)	206.9 (12.5)	195.2 (9.9)	186.3 (10.7)
Loco. stereotypy	3.3 (2.1)	11.2 (4.5)	14.0 (7.1)	21.1 (7.0)

All values represent average duration (sec) from 5-min testing sessions. Standard errors in parentheses.

Cortisol Response to Separation

There was a significant effect of Time Point on cortisol values [$F(14,45)=147.48$, $p<.001$]. Values one and two hours immediately following separation were elevated over Basal and Day 4 values ($p<.01$ for both comparisons). In addition, Basal cortisol was higher than Day 4 cortisol, and Hour 2 values were elevated over Hour 1 levels ($p<.05$ in both cases). The values at each time point are shown in Figure 1.

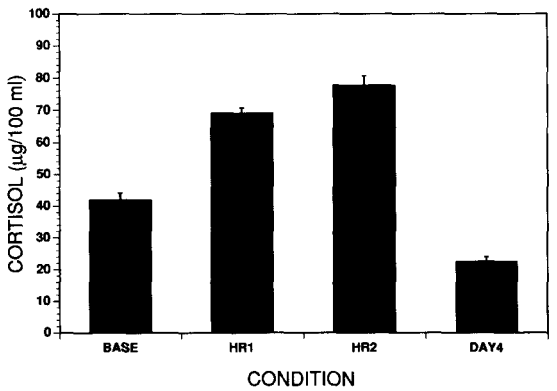


Fig. 1. Average plasma cortisol levels (µg/100ml) of rhesus monkey mothers during a series of separations from their 5-month old infants. BASE: The average of two basal samples: Preseparation (two weeks prior to separation, concurrent with CSF sampling) and Postseparation (two weeks following the fourth separation, concurrent with CSF sampling); HR1: one hour following separation (averaged across four weeks of separations); HR2: two hours following separation (averaged across four weeks of separations); DAY4: average of samples taken on Day 4 of Weeks 1 and 4, concurrent with CSF sampling.

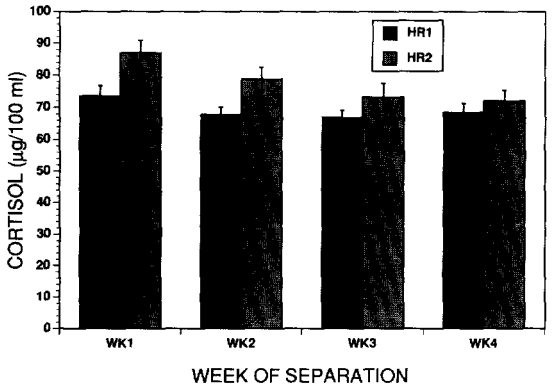


Fig. 2. Average plasma cortisol levels (μg/100ml) of rhesus monkey mothers during a series of separations from their 5-month old infants, one and two hours postseparation, across four consecutive weeks of separation. WK1: First week of separation; WK2: second week of separation; WK3: third week of separation; WK4: fourth week of separation; HR1: one hour following separation; HR2: two hours following separation.

When the Hour 1 and Hour 2 postseparation cortisol values were analyzed separately, there were effects of both Week of Separation [$F(3,42)=4.69$, $p<.01$] and Hour Postseparation [$F(1,14)=16.77$, $p<.001$]. As depicted in Figure 2, Week 1 values were higher than those in Weeks 2 through 4 ($p<.05$ for Week 1 – Week 2 comparison; $p<.01$ for Week 1 – Week 3 and Week 1 – Week 4 comparisons). In addition, the mean Hour 2 value was higher than that of Hour 1.

Analysis of only those samples obtained during Weeks 1 and 4 revealed a significant Week × Sample Condition interaction, $F(2,28)=8.19$, $p<.005$. This interaction could be accounted for by a significant difference ($p<.05$) between the Hour 2 samples; the Week 1 Hour 2 sample was significantly higher than the Week 4 Hour 2 sample. There were no differences between weeks for either the Hour 1 or Day 4 samples.

Preseparation cortisol values were significantly higher than Postseparation values $t(14)=2.67$, $p<.05$; the mean Preseparation value was 46.6 (± 2.33), whereas the average Postseparation value was 37.14 (± 3.33).

BODY WEIGHT

There was a significant main effect of Time Point [$F(2,28)=60.62$, $p<.001$]; body weight declined an average of 13% between the Preseparation period and Separation 4 ($p<.01$). The monkeys regained weight in the two-week Postseparation period ($p<.01$); however, the Postseparation weights were still significantly lower than the Preseparation weights ($p<.01$ by Duncan analysis). The average Preseparation weight was 6.71 kg (SEM=0.3), that of Separation 4 was 5.82 kg (SEM=0.32), and for Postseparation the average weight was 6.24 kg (SEM=0.33).

DISCUSSION

The results indicate that infant separation and removal to an unfamiliar environment constitutes a potent stressor for rhesus mothers. Infant separation in a novel environment initially elicited behavioral agitation and cortisol elevation. The acute behavioral response

to separation diminished across successive days postseparation. Repetitive separations continued to evoke behavioral protest and adrenocortical elevations; nonetheless, some habituation in both the behavioral and the adrenocortical responses was observed across separation experiences.

The biphasic behavioral response of the rhesus mother to infant separation resembles that reported for rhesus infants (SUOMI et al., 1983): vocalizations and locomotion were higher on the first day of separation, and self-directed behaviors and environmental exploration peaked during Days 2–4. Increases in self-directed behaviors from the acute to the chronic phase of separation have been construed as a sign of behavioral despair (MINEKA & SUOMI, 1978). However, our data indicate that these increases in self-directed behavior represent the resumption of normal self-maintenance behaviors that were suppressed during the first day of separation. The animals in the present study rarely exhibited any of the other classical signs of despair such as huddling or rocking.

For dyad-housed rhesus mothers, separation into an unfamiliar environment evokes more active behavior than that observed when the separated mothers remain in the home cage. This can be attributed to the combined effects of separation, social isolation, and novelty. SUOMI et al. (1983) reported that separated mothers which remained in their home cages exhibited mostly passive behavior throughout the separation. In contrast, the mothers in the present study exhibited less passive behavior, and more stereotyped locomotion and environmental exploration, than the mothers in the SUOMI et al. (1983) study. This is consistent with findings that, in rhesus infants, separation into an unfamiliar environment evokes more intense protest behavior than separation in the home cage (HINDE & DAVIES, 1972; HINDE & MCGINNIS, 1977). However, the amount of vocalization was approximately equal in the present study and the SUOMI et al. (1983) study, which implies that presence or absence of infant, rather than familiarity or novelty of the environment, is the factor determining the rate of vocalizations.

The finding of slight increases in stereotyped locomotion and decreases in passive behavior in rhesus mothers across successive weeks of separation stands in contrast to all published accounts of infant and adolescent rhesus, in which protest behaviors such as locomotion decrease, and despair behaviors such as self-clasp, rock, and huddle increase, across repeated separations (SUOMI et al., 1970; MINEKA et al., 1981; SUOMI et al., 1983). Stereotyped locomotion, far from being an "abnormal" behavior, probably is an expression of the motivation to search for the missing infant. Why mothers would continue to increase attempts to contact infants whereas younger animals relinquish their protest behaviors and exhibit increased signs of depression across successive weeks of separation is unclear.

Although clear elevations in cortisol at Hours 1 and 2 over the average basal (mean of Preseparation and Postseparation) values were observed, a few caveats should be noted. Our procedure confounded the effects of ketamine anesthesia and separation: the Preseparation and Postseparation blood samples were obtained under anesthesia whereas the Hour 1 and Hour 2 samples were not. In addition, the blood samples obtained under anesthesia were obtained much later following initial disturbance than those obtained one and two hours postseparation (e.g. 35 min average elapsed time for Preseparation samples vs five min average elapsed time for Hour 1 and Hour 2 samples). It is obvious that neither the Preseparation nor the Postseparation cortisol values reflect normative basal values for adult female rhesus macaques (in the range of 7–30μg/ml, depending on time of day and housing condition: LESHNER et al., 1978; SOCOL et al., 1978; REINHARDT et al., 1990). This may be due to the abnormally long time required to obtain the sample in these conditions.

It should also be noted that, unlike in the other sampling conditions, during Preseparation there was a significant correlation ($r=0.61$, $p<.05$) between time of sampling from initial room entry and cortisol values, whereas during Postseparation there was a positive correlation between order of sampling and cortisol values ($r=0.62$, $p<.05$).

The Day 4 cortisol values are more consistent with basal levels as reported in the literature. We believe that these two data points are valid despite the confound of sedation and somewhat longer elapsed time to sample (average 15 min). Studies have demonstrated that, in the short term, ketamine does not elevate cortisol levels above those typically found in response to handling and restraint (e.g. ELVIDGE et al., 1976; FULLER et al., 1984; WALKER et al., 1987), although the long term response to ketamine anesthesia may be a rise in cortisol (CROCKETT et al., 1993). It is possible that the delays associated with sampling during the Preseparation and Postseparation conditions resulted in a diminution of the inhibition of cortisol increases. Alternatively, it is possible that emergence from ketamine, with its attendant disorientation and ataxia, can trigger cortisol increases. It may be that the animals in the Preseparation and Postseparation conditions were already beginning to emerge from anesthesia by the time their blood was sampled, hence resuming their cortisol reactivity.

It could be hypothesized that the cortisol elevations observed at one and two hours postseparation were simply artifacts of the separation and handling procedures. Although this may have contributed to the observed effects, it does not entirely account for the large increases in cortisol. It has been demonstrated that separation, handling, and returning to the home cage results in only transient cortisol elevations, at least in squirrel monkey mothers (JORDAN et al., 1985). In addition, previous work showed that the effects of handling and a separation stressor on cortisol levels are additive (CHAMPOUX et al., 1989).

In this study, cortisol elevation occurred reliably in response to repeated separations. These findings were consistent with those reported in squirrel monkey infants and mothers (COE, GLASS, WIENER, & LEVINE, 1983; HENNESSY, 1986). Nonetheless, cortisol values decreased slightly across consecutive weeks of separation, indicating that there was some degree of response habituation. This stands in contrast to the separation response of squirrel monkey mothers (COE, GLASS, WIENER, & LEVINE, 1983). These contrasting findings may be artifacts of the separation procedures utilized (COE, GLASS, WIENER, & LEVINE, separated 3-month old infants six times for one hour each), or may reflect a species difference in adrenocortical reactivity.

These findings demonstrate that, under certain conditions, the rhesus mother is capable of exhibiting behavioral and adrenocortical perturbations as a consequence of infant separation. The extent to which the separation experience, social isolation, and exposure to novelty contribute to this effect remains to be tested. These three factors are in likelihood additive. For instance, novel environments can be stressful even in the absence of a separation. MITCHELL and GOMBER (1976) and CROCKETT et al. (1993) have demonstrated that moving rhesus monkeys into novel cages evoked behavioral agitation and cortisol elevation; however, these are small and transient effects (CROCKETT et al., 1993). In addition, in the rhesus macaque, social isolation is itself associated with elevated cortisol levels (SACKETT et al., 1973; CHAMPOUX et al., 1989).

In infant rhesus, the social separation paradigm has been utilized as a model of depression (SUOMI & HARLOW, 1977), as a measure of mother-infant attachment (HINDE & SPENCER-BOOTH, 1971), and as an assessment of individual differences in stress reactivity (SUOMI, 1983). The results of the present study indicate that the social separation procedure might be an effective assessment tool for use with rhesus mothers as well.

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